Aidar Alimbayev

PhD Candidate in Machine Learning, MBZUAI

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Research Summary

PhD candidate building physics-aware machine learning for atomistic systems: transferring pretrained machine-learning interatomic potentials (UMA) to electronic-structure targets, electron-density models for molecular property prediction, and SE(3)-aware attention for molecular transformers. Seeking a **Summer 2027 research internship** on MLIPs, atomistic foundation models, or molecular and materials property prediction.

Interests: MLIPs and atomistic foundation models · molecular and materials property prediction · equivariant GNNs and transformers · electronic structure (DOS, electron density) · generative materials design

Education

PhD in Machine Learning — Aug 2024 – present
Mohamed bin Zayed University of Artificial Intelligence (MBZUAI) — Abu Dhabi, UAE
Thesis: *Physics-Aware Foundation Model for Atomistic Systems* · Supervisor: Prof. Martin Takáč

MSc in Engineering and Technology, Radio Communication Technologies — 2015 – 2017
Bonch-Bruevich St. Petersburg State University of Telecommunications — St. Petersburg, Russia

BSc in Engineering and Technology, Radio Engineering, Electronics and Telecommunications — 2010 – 2015
S. Seifullin Kazakh Agro Technical University — Astana, Kazakhstan

Research Experience

PhD Researcher (supervisor: Prof. Martin Takáč) — Aug 2024 – present
MBZUAI — Abu Dhabi, UAE

- **UMA-DOS.** Fine-tuned pretrained UMA MLIP backbones with a shape–amplitude spectral decoder to predict the electronic density of states on Materials Project; best controlled nMAE (0.112 vs. 0.135, −17%) and spectral-angle error among reproduced baselines. Showed that end-to-end fine-tuning beats both frozen transfer and scratch training.
- **QDFaiNN** (first author). Added a PaiNN-style directional block and a learned pseudo-electronegativity that deform Gaussian basis functions inside a Quantum Deep Field model; −34% / −32% interpolation / size-extrapolation error on QM9 and QM7-X for only +3.6% parameters.
- **LAMP-QDF.** Replaced the pointwise density-to-potential (Hohenberg–Kohn) map with distance-aware local graph attention; QM9 test MAE 1.06 kcal/mol (−26%) and −28.5% error on larger, unseen molecules.
- **Molecular transformers.** Controlled study of attention formulations under fixed SE(3)-aware positional encodings (QM9, Molecule3D); identified the best injection and gating scheme (−15% normalised error).
- **MOF inverse design** (MFIRP: IIT Delhi, IIT Delhi–Abu Dhabi, MBZUAI). UMA encoder with a flow-matching decoder conditioned on target CO_2 uptake. Re-engineered the data pipeline for 324,426 MOF structures (Gemmi parsing 50×, MatGL graph construction 76×faster); reproduced CDVAE above its published Carbon-24 match rate (65.7% vs. 55.2%).
- Collaborations on latent-diffusion inverse problems (PASEO: up to 19×faster, 34% less memory) and Kolmogorov–Arnold modules for latent ODE models (−98.7% parameters at equal accuracy).

Visiting Scholar (host: Prof. Ulisses Braga-Neto) — Aug – Dec 2022
Texas A&M University, Dept. of Electrical and Computer Engineering — College Station, USA

- Physics-informed neural networks for complex systems; stacked generalisation with particle swarm optimisation. Funded by the Shakhmardan Yessenov Foundation.

Research Assistant (supervisors: Prof. A. Gaipov, Dr. A. Zollanvari) — Aug 2021 – Dec 2022
Nazarbayev University School of Medicine, MES research grant — Astana, Kazakhstan

- Machine learning on Kazakhstan's national healthcare big data (UNEHS); first-author paper in *Scientific Reports*.

Leading Researcher — May 2016 – Jun 2017
Bonch-Bruevich St. Petersburg State University of Telecommunications — St. Petersburg, Russia

- Built automated test stands (NI DAQ, LabVIEW); ran workshops and professional training courses.

Publications

2026 papers and manuscripts

[1] K. Ziu, A. V. Cohen, Y. Ito, **A. Alimbayev**, L. Kronik, M. Takáč. UMA-DOS: Transfer Learning from Pretrained Atomistic Models for Density of States Prediction. *NeurIPS 2026*, submitted.

[2] D. Kumar, Y. Ito, **A. Alimbayev**, K. Ziu, M. Takáč. LAMP-QDF: Local Attention Map for Potential in Quantum Deep Field. *SimBioChem Workshop, NeurIPS 2026*, **accepted**.

[3] A. Shtanchaev, A. Asadulaev, **A. Alimbayev**, K. Salta, M. Takáč, E. Moulines. Perturb-and-Solve: Efficient Learned-Operator Conditioning for Latent Diffusion Inverse Problems. *BeNTo Workshop, NeurIPS 2026*, **accepted**; *ICLR 2027*, under review.

[4] Y. Ito, **A. Alimbayev**, K. Ziu, D. Kumar, K. Ono, M. Takáč. Revisiting Attention Formulations for Molecular Transformers with Positional Encoding. *TMLR*, under review.

[5] **A. Alimbayev**, Y. Ito, D. Kumar, K. Ziu, A. Shtanchaev, A. V. Cohen, L. Kronik, M. Takáč. Variational Pseudo Electronegativity: an Orbital-Resolved Basis Deformation inside a Quantum Deep Field Model. Manuscript in revision.

[6] R. Gao, D. Kumar, **A. Alimbayev**, K. Ziu, Y. Ito, A. Shtanchaev, R. Bresson, M. Takáč. Kolmogorov–Arnold Networks for Compact and Interpretable Latent Linear ODE Models. *Sim2Sci Workshop, NeurIPS 2026*, submitted.

Earlier journal and conference papers

[7] **A. Alimbayev**, G. Zhakhina, A. Gusmanov, Y. Sakko, S. Yerdessov, I. Arupzhanov, A. Kashkynbayev, A. Zollanvari, A. Gaipov. Predicting 1-year mortality of patients with diabetes mellitus in Kazakhstan based on administrative health data using machine learning. *Scientific Reports*, 13, 8412 (2023).

[8] A. Gusmanov, G. Zhakhina, S. Yerdessov, Y. Sakko, K. Mussina, **A. Alimbayev**, D. Syssoyev, A. Sarria-Santamera, A. Gaipov. Review of the research databases on population-based registries of Unified electronic Healthcare system of Kazakhstan (UNEHS). *International Journal of Medical Informatics*, 170, 104950 (2023).

[9] A. Mirmanov, **A. Alimbayev**, S. Baiguanysh, N. Nabiev, A. Sharipov, A. Kokcholokov, D. Caratelli. Development of an IoT platform for stress-free monitoring of cattle productivity in precision animal husbandry. *ASTES Journal*, 6(1), 501–508 (2021).

[10] **A. Alimbayev**, V. S. Uss, A. B. Mirmanov, P. P. Shumakov. Test stand for prospective mobile power-supply sources. *IEEE SIBCON* (2017).

[11] S. B. Baiguanysh, A. B. Mirmanov, **A. Alimbayev**, A. Y. Makartseva. Implementation of PPM modulation and demodulation using NI Digital Electronics FPGA board. *IEEE SIBCON* (2015).

Technical Skills

Deep learning	PyTorch; equivariant message passing (PaiNN-style); transformers and attention design; VAEs, diffusion and flow matching; Kolmogorov–Arnold networks
Atomistic ML	UMA / fairchem MLIP backbones, fine-tuning and transfer; MatGL, pymatgen, Gemmi; CDVAE; QM9, QM7-X, Molecule3D, Materials Project, MOF databases
Physics	DFT and Kohn–Sham basics, Hohenberg–Kohn theorem, LCAO and Gaussian basis sets (6-31G), electron density, density of states, SE(3) symmetry
Computing	Python, C/C++, MATLAB, Git, LaTeX; Slurm GPU clusters (HPC)
Hardware	LabVIEW and NI DAQ, embedded systems (STM32, AVR), Altium Designer, Fusion 360

Teaching, Engineering and Leadership

Co-instructor, AI1215 Introduction to Machine Learning (business track) — Summer 2026
MBZUAI (with Prof. Zangir Iklassov); designed the from-scratch Python labs — Abu Dhabi, UAE

Engineer, Fab Lab NU; supervisor of student engineering teams — Jun 2018 – Jul 2024
NURIS, Nazarbayev University — Astana, Kazakhstan

- Built research test stands (offshore-structure fatigue, shape-memory composites, food drying) and set up the electronics lab.
- Supervised Shell Eco-marathon teams (Singapore, Indonesia, Qatar; 2018–2025) and the team that won 1st place in two categories at the Castellolí Smart Scooter Challenge 2019 (Spain).

Project team member, ElectroLogic innovation project — Nov 2023 – Aug 2024
InPro-Tech LLP (QazInnovations commercialisation grant) — Astana, Kazakhstan

- Developed and piloted a 12-channel programmable logic controller for building energy efficiency; 10 units installed at Nazarbayev University.

Teaching Assistant (digital signal processing, LabVIEW); earlier Laboratory Manager — 2013 – 2015, 2017 – 2018
S. Seifullin Kazakh Agro Technical University — Astana, Kazakhstan

Honours and Service

- Shakhmardan Yessenov Foundation fellowship for a visiting scholarship at Texas A&M University (2022).
- Grant committee member (volunteer), Shakhmardan Yessenov Foundation international internships competition (2024 – present).

Languages

Kazakh (native) · Russian (native) · English (fluent, IELTS 6.5) · Spanish (beginner)